SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: **MLA03**
COURSE NAME: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr DUMPALA PRASANTH

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Data Interpretation **Duration: 1hr**

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.
(5 Marks)

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. **(5 Marks)**

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. **(5 Marks)**

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

7. A Q-learning agent is tested with different exploration strategies: ε-greedy, softmax, and random exploration. The cumulative reward is highest with ε-greedy. Compare the strategies and defend why ε-greedy achieves better performance in this scenario. **(5 Marks)**

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **(5 Marks)**

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective. **(5 Marks)**

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning. **(5 Marks)**

Code of Conduct:

I ___K Uday Kiran Reddy (192425211)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: k Uday

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

1. Interpretation of Increasing Rewards

Understanding of Data

The reward values increase gradually from **10 to 45** over multiple training episodes. This represents a clear positive trend in the agent's learning performance. The increasing rewards indicate that the agent is progressively making better decisions and receiving higher returns from the environment.

Analysis

At the beginning of training, the agent has limited knowledge about the environment, so its actions may be inefficient and result in lower rewards. As more episodes are completed, the agent updates its knowledge and learns which actions produce better outcomes. The gradual increase from 10 to 45 suggests that the learned policy is improving consistently rather than randomly.

A sustained increase in reward is also an indication that the agent is reducing poor decisions and selecting more beneficial actions. If the reward eventually becomes stable at a high value, it would provide stronger evidence that the policy is approaching convergence.

Application of Concept

In reinforcement learning, the objective is to learn a policy that maximizes expected cumulative reward. The observed reward improvement demonstrates the effect of repeated interaction, learning, and policy refinement.

Conclusion

The increase in reward from **10 to 45 indicates significant learning progress and policy improvement**. The agent is becoming more effective at selecting actions. However, the data alone does not prove that the policy is globally optimal; additional evidence such as reward stabilization or convergence would be required.

2. Effect of Epsilon on Cumulative Reward

Understanding of Data

The epsilon values are **0.1, 0.3, and 0.5**. The highest cumulative reward is obtained when **ε = 0.1**, while the lowest reward occurs at **ε = 0.5**.

Analysis

Epsilon (ε) controls the exploration-exploitation trade-off in the epsilon-greedy strategy.

- **ε = 0.1:** The agent explores only occasionally and mostly exploits the best-known action.
- **ε = 0.3:** The agent performs more exploration and consequently makes more exploratory decisions.
- **ε = 0.5:** Half of the action selections involve exploration, resulting in substantially less exploitation of the currently learned best actions.

The observed highest reward at $\varepsilon = 0.1$ indicates that the environment benefits from exploiting learned knowledge. At $\varepsilon = 0.5$, excessive exploration causes the agent to select less beneficial actions more frequently, reducing cumulative reward.

Application of Concept

The epsilon-greedy policy selects a random action with probability ε and the currently best-known action with probability $\mathbf{1 - \varepsilon}$. Therefore, lower epsilon values favor exploitation, whereas higher epsilon values favor exploration.

Conclusion

ε = 0.1 provides the best performance among the tested values because it maintains limited exploration while allowing the agent to exploit its learned policy effectively. The result demonstrates that an appropriate balance between exploration and exploitation is essential for maximizing reward.

3. Q-Learning vs SARSA

Understanding of Data

Q-Learning and SARSA are compared using average rewards over **100, 200, and 300 episodes**. Q-Learning consistently obtains slightly higher rewards than SARSA.

Analysis

The consistent performance advantage of Q-Learning indicates that it is learning a more rewarding action policy under the observed conditions.

Q-Learning is an **off-policy** algorithm. It updates its Q-value using the maximum estimated value of the next state:

$$\mathbf{Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max Q(s',a') - Q(s,a)]}$$

SARSA is an **on-policy** algorithm and updates according to the action that the current policy actually selects:

$$\mathbf{Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma Q(s',a') - Q(s,a)]}$$

Because Q-Learning directly considers the maximum future Q-value, it can learn an aggressive reward-maximizing policy. SARSA considers the actual next action, including the effects of its exploration strategy.

Application of Concept

Since Q-Learning produces slightly higher rewards at all three episode levels, the observed data indicates better performance for Q-Learning in this particular experiment.

Conclusion

Q-Learning performs better based on the observed reward values. However, the question provides reward comparisons rather than explicit convergence points, so it is more accurate to say that Q-Learning **shows faster or stronger convergence based on the observed trend**, rather than claiming mathematically proven faster convergence.

4. Effect of Discount Factor

Understanding of Data

The discount factors tested are **0.2, 0.5, and 0.9**. The average reward increases as the discount factor increases.

Analysis

The discount factor, represented by **γ**, determines how strongly future rewards influence the agent's current decision.

- **γ = 0.2:** The agent strongly favors immediate rewards.
- **γ = 0.5:** The agent considers both immediate and future rewards.
- **γ = 0.9:** The agent gives substantial importance to future rewards.

The observed increase in average reward indicates that considering future rewards is beneficial in this environment. A high discount factor encourages the agent to select actions that may provide smaller immediate rewards but produce greater long-term returns.

Application of Concept

The Q-learning target contains the discounted future reward:

$$\mathbf{r + \gamma \max Q(s',a')}$$

Therefore, increasing γ increases the contribution of future rewards to the learning process.

Conclusion

Among the tested values, **γ = 0.9 is the most suitable** because it produces the highest average reward. This suggests that long-term planning is important for achieving better learning performance in the given environment.

5. Improvement in Success Rate

Understanding of Data

The success rate increases from **45% at 50 episodes to 92% at 200 episodes**.

This represents an improvement of:

92% − 45% = 47 percentage points

The success rate more than doubles relative to the initial 45% level.

Analysis

The significant improvement indicates that the agent is learning from repeated interactions with the environment. At 50 episodes, the policy is still relatively immature, resulting in many unsuccessful decisions. By 200 episodes, the agent has learned substantially better action choices.

A success rate of 92% indicates strong policy performance. However, an optimal policy cannot be conclusively established solely from the given data because there is no information about the theoretical maximum success rate or whether the success rate has converged.

Application of Concept

An RL agent improves its policy by updating its knowledge based on rewards and experiences. The increasing success rate is therefore evidence of effective learning and policy refinement.

Conclusion

The increase from **45% to 92% strongly indicates substantial policy improvement**. The agent appears to be approaching an effective policy, but the available data does not definitively prove that it has learned the globally optimal policy.

6. Effect of Learning Rate in CartPole

Understanding of Data

The learning rates tested are **0.01, 0.05, and 0.1**. The agent achieves higher stability at **0.05** compared with the other values.

Analysis

The learning rate, represented by **α**, determines how strongly new experience changes the agent's learned values.

- **α = 0.01:** Learning is slow because updates are small. This may provide stability but can result in slow convergence.
- **α = 0.05:** Provides a moderate update size, allowing the agent to learn sufficiently quickly while maintaining stability.
- **α = 0.1:** Updates are larger, which can make learning faster but may produce instability or oscillations.

The observed stability at 0.05 indicates that this value provides an effective balance between learning speed and stability.

Application of Concept

The learning-rate parameter determines the magnitude of updates to learned values. An appropriate learning rate prevents the agent from adapting either too slowly or too aggressively.

Conclusion

A learning rate of 0.05 provides the best trade-off among the tested values. It allows meaningful learning while maintaining greater stability, making it the most suitable choice for the given CartPole experiment.

7. Comparison of Exploration Strategies

Understanding of Data

The Q-learning agent is tested using **epsilon-greedy, softmax, and random exploration**. The cumulative reward is highest with epsilon-greedy exploration.

Analysis

The three strategies behave differently:

Epsilon-greedy:
Usually selects the best-known action while occasionally exploring another action. This provides a simple balance between exploitation and exploration.

Softmax:
Selects actions probabilistically according to their estimated values. Better actions receive higher selection probabilities, but lower-valued actions can still be explored.

Random exploration:
Selects actions without strongly considering their estimated values. This can cause the agent to frequently select poor actions.

The highest cumulative reward with epsilon-greedy indicates that its balance between exploitation and exploration is most effective under the conditions of this experiment.

Application of Concept

Epsilon-greedy allows the agent to exploit known high-value actions while retaining some exploration. This helps the agent discover alternatives without sacrificing too much reward.

Conclusion

Epsilon-greedy achieves the best performance in the given scenario because it provides an effective and simple exploration-exploitation balance. The result is specific to the observed experiment; different environments could favor another strategy.

8. Reduction in Steps in Maze Environment

Understanding of Data

Initially, the agent requires **50 steps** to reach the goal. After 300 episodes, it requires only **15 steps**.

The reduction is:

50 − 15 = 35 steps

This represents a **70% reduction** in the number of steps:

(35 / 50) × 100 = 70%

Analysis

Fewer steps indicate that the agent has learned a more efficient path through the maze. Initially, the agent likely explores different paths and makes inefficient decisions. With repeated training, it learns which actions lead more directly toward the goal.

The reduction to 15 steps demonstrates significant improvement in policy efficiency. However, whether 15 steps represents the optimal policy depends on the shortest possible path in the maze, which is not provided.

Application of Concept

In a goal-based RL environment, a policy can be evaluated based on the number of steps required to achieve the goal. A reduction in steps indicates that the learned policy is becoming more efficient.

Conclusion

The reduction from **50 to 15 steps strongly indicates policy improvement and learning convergence**. It suggests that the agent is approaching an efficient policy, but optimality cannot be conclusively established without knowing the minimum possible number of steps.

9. Effect of Discount Factor in SARSA

Understanding of Data

The SARSA agent is tested with discount factors **0.4, 0.7, and 0.95**. The highest long-term reward is achieved with **$\gamma = 0.95$**.

Analysis

A discount factor determines how much importance the agent gives to future rewards.

- **$\gamma = 0.4$:** The agent places greater emphasis on immediate rewards.
- **$\gamma = 0.7$:** The agent considers future rewards to a moderate extent.
- **$\gamma = 0.95$:** The agent strongly considers future rewards.

Since $\gamma = 0.95$ produces the highest long-term reward, the environment appears to benefit from long-term planning. The agent is able to accept actions whose immediate reward may be smaller when those actions lead to greater future returns.

Application of Concept

The SARSA update is:

$$\mathbf{Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma Q(s',a') - Q(s,a)]}$$

With $\gamma = 0.95$, the future Q-value contributes strongly to the target. Therefore, the agent learns to consider the consequences of its current action over a longer time horizon.

Conclusion

$\gamma = 0.95$ is the most effective value among those tested because it produces the highest long-term reward. This demonstrates that future rewards are particularly important for successful decision-making in the given environment.

10. Stabilization of Rewards in a Stochastic Environment

Understanding of Data

The RL agent experiences **fluctuating rewards during early episodes**, but the rewards stabilize after approximately **500 episodes**.

Analysis

Reward fluctuations during early training are expected because the agent has limited knowledge and is actively exploring the environment. In a stochastic environment, randomness can also cause different outcomes even when similar actions are selected.

As training continues, the agent learns from its experiences and improves its policy. Stabilization after 500 episodes indicates that the agent's decisions have become more consistent and that the learned policy is producing relatively stable outcomes.

The stabilization is especially important in a stochastic environment because complete elimination of reward variation may not be possible. Therefore, a stable reward trend rather than perfectly constant rewards is a meaningful indicator of learning robustness.

Application of Concept

RL algorithms continuously update their value estimates and policies using experience. With sufficient training, these estimates can become more stable, resulting in a more consistent policy.

Conclusion

The transition from fluctuating rewards to stable rewards after **500 episodes indicates significant policy improvement and learning robustness**. It suggests that the agent has learned a reliable strategy despite environmental uncertainty. Stable performance provides evidence of convergence, although proving an optimal policy would require additional information such as the optimal expected reward.